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DiffraScout: An auditable Python workflow for generating powder diffraction references

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Summary

Researchers compare measured powder X-ray diffraction (PXRD) patterns with reflections calculated from candidate crystal structures. Preparing these references requires the structure setting, radiation, and calculation parameters to remain linked. DiffraScout is an open-source Python package that converts composition queries or local Crystallographic Information Files (CIFs) into indexed theoretical reflections and display profiles (Gong, 2026). Its command-line, Python, and desktop interfaces write source structures, settings, data, diagnostics, and tables to a result bundle with a verifiable manifest. The formatted workbook presents an overview and can be previewed before saving. DiffraScout supports experimental planning and interpretation; it does not identify phases in measured patterns or refine experimental data.

Statement of need

Powder-diffraction screening links candidate compositions and structures to expected reflection positions and intensities. Preparing references spans alloy parsing, database queries, CIF validation, diffraction calculations, optional tensor matching, and spreadsheet transfer. Across separate tools, candidate identifiers, structure settings, radiation, warnings, and calculation parameters can become detached from the final table.

DiffraScout connects these steps for materials researchers and students preparing references before or alongside experiments. Runs start from a composition, provider ID, or local CIF; the software records source structures and data interpretation, then exports machine-readable tables and a reviewable workbook in one bundle. Local-CIF processing needs no database credentials. This workflow makes theoretical references repeatable while leaving experimental interpretation to the researcher.

State of the field

The Materials Project provides computed structures and properties (Jain et al., 2013), and pymatgen supplies materials-analysis objects and calculations (Ong et al., 2013). Gemmi parses CIFs and performs crystallographic operations and structure-factor calculations (Wojdyr, 2022); spglib offers an independent symmetry search (Togo et al., 2024). Dans_Diffraction simulates diffraction from supplied structures (Porter & Prestipino, 2023), and Elasticity analyzes linear elasticity and tensors (Depriester & Kubler, 2025). For measured data, GSAS-II supports reduction and refinement (Toby & Von Dreele, 2013), while pyFAI provides detector calibration and azimuthal integration (Ashiotis et al., 2015).

DiffraScout introduces no new diffraction or elasticity algorithm. Instead, it adds an

39 application layer that links database or local-CIF inputs, candidate identities, user-selected
40 settings, calculations, diagnostics, and exports in one result bundle—relations that extend
41 beyond any one numerical library. It restructures workflow components from PhaseScout and
42 CIF2Peaks (D-sudoasd, 2026b, 2026a) around shared records and provenance. Database access
43 is optional, so local-CIF analysis runs offline.

44 Software design

45 A provider supplies normalized candidate records and their source artifacts. The Materials
46 Project adapter records query context and source identifiers; the local-CIF path requires
47 no network service. Candidates are deduplicated before download. Each structure record
48 preserves the CIF hash, selected data block, cell, space-group resolution, occupancy state, and
49 diagnostics. Conflicting space-group declarations and partial occupancies remain visible for
50 review rather than being silently repaired (Hall et al., 1991).

51 The diffraction engine uses Gemmi to enumerate symmetry-related reflection families, apply
52 systematic-absence rules, and calculate X-ray structure factors. Results include representative
53 Miller indices, interplanar spacing, scattering-vector magnitude, multiplicity, and separately
54 identified intensity channels with and without the stated Lorentz–polarization correction. A
55 Gaussian, Lorentzian, or pseudo-Voigt profile is generated for display; it is not an instrument-
56 response fit. Configured limits on profile-grid points and reciprocal-space candidate counts are
57 checked before allocation.

58 Directional Young's moduli can be attached when an elastic tensor is uniquely associated
59 with the CIF and its coordinate frame is usable. The Materials Project's elastic constants
60 are computed data, not experimental measurements (Materials Project, n.d.). DiffractScout
61 validates the matrix, unit, conditioning, and positive definiteness before evaluation along a
62 reflection-plane normal. When a downloaded tensor lacks a verified transformation into the
63 CIF Cartesian frame, the software retains its provenance and emits a frame-transformation
64 status without a directional value. This fail-closed policy avoids making a numerically plausible
65 but unsupported tensor-to-structure pairing.

66 The command-line and desktop interfaces call the same analysis pipeline. The command-line
67 interface accepts reusable parameter presets and can inspect and verify an existing result
68 bundle without rerunning the analysis. The desktop application offers separate result preview,
69 workbook saving, and result-folder actions; versioned parameter presets exclude credentials
70 and file paths. The Excel exporter formats existing result values rather than recalculating
71 them. It adds an overview, readable number formats, filters, frozen phase-identity columns,
72 and grouped technical fields while preserving the corresponding tabular records. CSV and
73 JSON outputs remain available for scripted use.

74 Each run is first written to a staging directory. Before publication to the requested output
75 path, a verifier checks the manifest's SHA-256 digests and file sizes, path safety, symbolic links,
76 and unlisted files. Replacing an existing result requires explicit authorization and verification
77 of that result. This design makes the completed directory a portable evidence bundle while
78 keeping recovery behavior explicit.

79 An offline analytic benchmark tests simple-cubic, body-centered cubic, face-centered cubic,
80 and NaCl selection rules, reflection geometry, structure factors, and cubic elasticity against
81 closed-form expectations. A separate pymatgen XRDCalculator comparison (Ong et al., 2013)
82 matched 40 d-spacing groups from four synthetic CIFs using Cu K α (1.5406 Å) over 5–120° 2 θ .
83 Every group met the specified spacing (10^{-6} Å) and angle (10^{-5}°) tolerances, and all tested
84 allowed and forbidden reflection rules passed. Directional moduli along [100], [110], and [111]
85 for a synthetic cubic tensor passed the defined absolute and relative tolerances in the pymatgen
86 compliance-tensor comparison (Ong et al., 2013). The automated suite checks analysis and
87 export contracts; continuous integration runs it on Linux for Python 3.10–3.13 and selected
88 checks on Windows. These comparisons are reproducible without a database service.

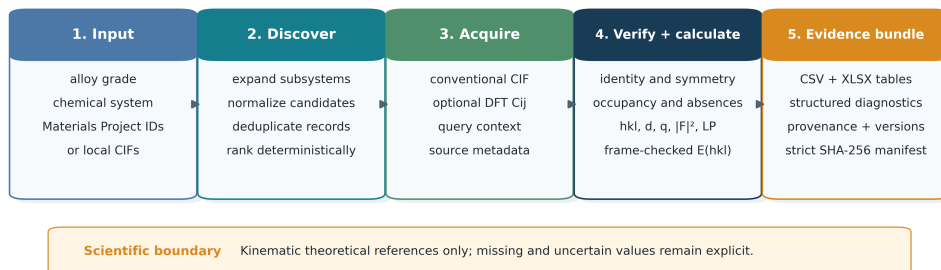


Figure 1: DiffractScout workflow from candidate request or local CIFs to a verified result bundle. Source identity, assumptions, diagnostics, and missing-data states are carried across the pipeline.

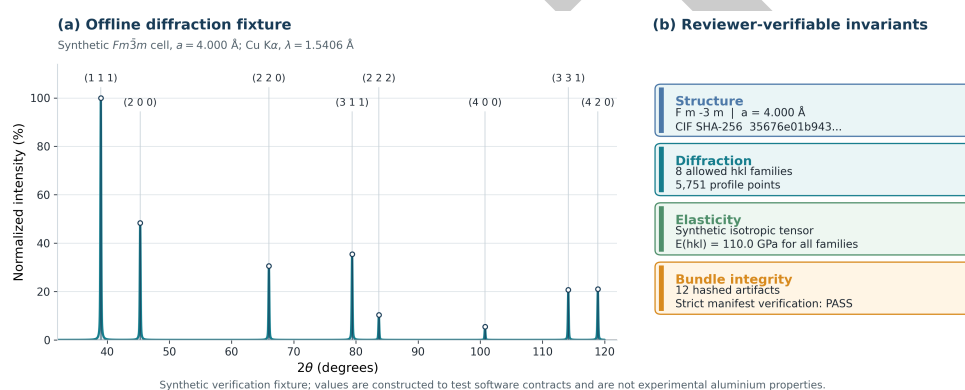


Figure 2: Analytic offline benchmark for the synthetic FCC fixture. The figure shows allowed reflection families and the selected geometry and bundle-integrity checks. The fixture is for software verification, not a measured material.

Research impact statement

The author reports using DiffractScout in the research underlying the Ti-24Nb-4Zr-8Sn study by Gong et al. (Gong et al., 2026), as confirmed on 27 September 2026. This is an author-confirmed application, not a claim that the article cites the software or that all features in the current candidate were used. The historical revision and operation-to-output mapping remain to be checked against the author's processing records before submission. The repository records this distinction alongside reproducible software-verification examples. Those examples establish specified numerical and software behavior rather than experimental accuracy or independent adoption.

AI usage disclosure

OpenAI GPT-5.6 Pro assisted earlier repository development, testing, documentation, figure production, and manuscript drafting from 30 July to 12 August 2026. In this revision, OpenAI Codex GPT-6 and cooperating assistant agents supported code changes, test scaffolding and review, documentation, citation discovery, manuscript restructuring, and automated verification. Software claims were checked against project source and scientific contracts, references against publisher or upstream records, and automated tests and the analytic benchmark were run. Human author review of the AI-assisted contributions remains pending.

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